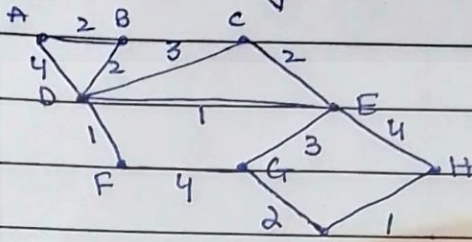


Kruskal's Algorithm For MST

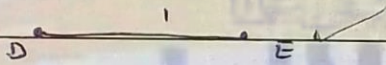
- i) List all edges of graph ✓
- ii) Sort edges according to their weight ✓
- iii) Add one by one edges to solution (excluding unsafe edges) ✓
- iv) Tree may not be connected During process.



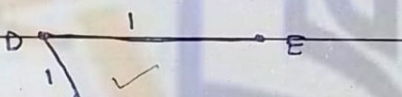
Edges: $(A,B:2), (A,D:4), (B,D:2), (C,D:3), (C,E:2), (D,E:1), (D,F:1), (E,G:3), (E,H:4), (F,G:4), (G,I:2), (H,I:1)$

Sort: $(D,E:1), (D,F:1), (H,I:1), (A,B:2), (B,D:2), (C,E:2), (G,I:2), (C,D:3), (E,G:3), (A,D:4), (E,H:4), (F,G:4)$

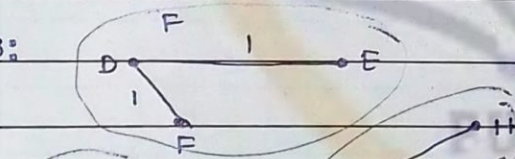
step 1:



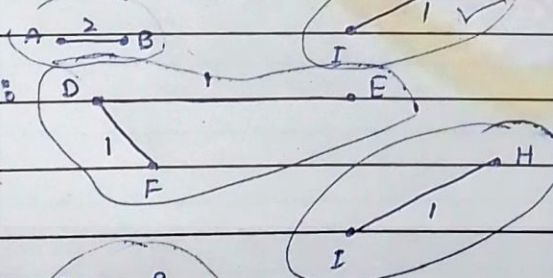
step 2:



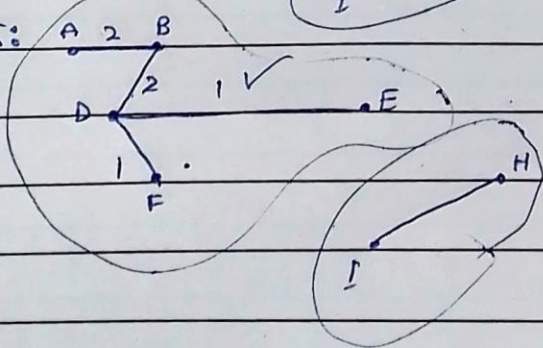
step 3:



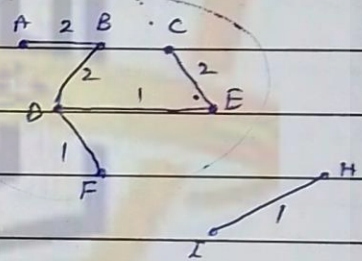
step 4:



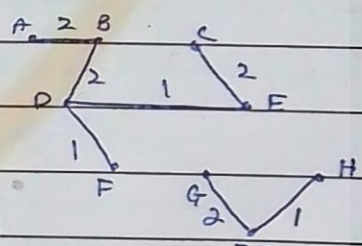
step 5:



step 6:



step 7:



step 8:

$(C,D:3)$ Not safe ✓
 $(E,G:3)$ safe.

